

Eye Tracking Test Suite: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Category: Research

ABSTRACT

Eye tracking is an increasingly popular method for interacting with augmented reality (AR) and measuring user attention. However, implementing and evaluating eye tracking across multiple platforms and use cases can be challenging due to the lack of standardized metrics and measurements. Additionally, existing calibration methods and accuracy measurements often do not account for the common scenarios of walking and scanning in mobile AR settings. To test and compare different eye tracking devices on various AR tasks and metrics, we developed EyeTTS, an eye tracking test suite specifically designed for scenarios involving head movement and locomotion in AR. We conducted user studies on the Magic Leap (n=36, 1 trial per task) and HoloLens 2 (n=54, 2 trials per task) devices to collect data and compare the precision and accuracy of each headset under different movement and reference frame conditions. Additionally, we performed post-hoc recalibration of the eye gaze data using the tasks in our test suite in order to study the impact of different calibration tasks on accuracy and compare them with the inbuilt calibration provided by the head-mounted display (HMD)s. Our analysis revealed lower eye tracking accuracy while walking, significant differences between the two tested headsets, and improvements to eye tracking performance while walking when a calibration technique involving stipulated head motion is employed.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies; Human-centered computing—Ubiquitous and mobile computing—Ubiquitous and mobile devices—Mobile devices;

1 INTRODUCTION

Eye tracking is a useful tool that has gained popularity in recent years. Its many applications, from diagnostics (gathering data on a user’s visual attention) to interaction (allowing users to use their gaze as a cursor) [8], make it a useful means of understanding user behavior and enhancing human-computer interaction. Recent advancements in eye-tracking technology have made it more accessible and affordable for a wider range of applications [12]. As such, it is important that developers and users understand the capabilities and limitations of their eye tracking devices during use.

With the rise of AR HMDs such as the HoloLens 2, Magic Leap, and the recently-announced Apple Vision Pro, eye trackers have been integrated into various platforms and use cases. However, implementing eye-tracking technology on multiple platforms and standardizing their performance is a difficult and complex problem. To address this challenge, specific metrics, such as accuracy and precision, are often selected to evaluate the performance of eye-tracking systems [7, 12, 17]. Although existing work has used these metrics to examine the accuracy of eye trackers while stationary, there has been little research into how locomotion affects eye tracking. Even the announcement for the Apple Vision Pro, which integrates eye tracking and interaction more than any previous AR headset, did not contain any examples of eye tracking used during locomotion [2]. It is clear that the effects of locomotion on eye tracking performance are underexplored and important to understand as AR use continues to grow.

Furthermore, current eye tracking device calibration methods rely on users keeping their heads stationary, without incorporating any head or body movement. It is unclear whether these existing eye tracking calibration methods are effective when a user intends to use eye tracking while walking. Therefore, it is important to investigate how effective existing eye tracking calibration methods are and potentially find new calibration procedures to incorporate locomotion.

In order to meet these needs, we developed EyeTTS, a test suite designed to assess eye tracking performance in various AR scenarios that involve movement and locomotion. Our test suite is presently compatible with the HoloLens 2 and the Magic Leap, enabling seamless data collection on those devices. The suite consists of several tasks that occur in different conditions: with the user’s head constrained to a chin rest, with the user’s body constrained through sitting, and with the user free to move and walk. In addition, these tasks present a stimulus in three reference frames: world, body, or screen stabilized. This test suite offers a comprehensive range of tasks that simulate various scenarios encountered in AR applications. These tasks facilitate the development of contextually informed designs for future eye trackers, enhancing their accuracy and precision. Additionally, the test suite provides a standardized approach for evaluating both user performance and device performance within these specific scenarios.

Using EyeTTS, we conducted two user studies to assess the spatial accuracy and precision of the gaze signal captured by the integrated eye tracker of the Magic Leap and HoloLens 2 headsets. We analyzed eye tracking performance in different movement conditions and identified a possible recalibration procedure that improves eye tracking accuracy during locomotion. Our analysis revealed that 1) eye tracking accuracy tended to decrease as head and body motion increased, 2) our two hitherto tested AR headsets yielded significantly different results on several of our EyeTTS tasks, and 3) introducing controlled head motion to a recalibration task resulted in better performance for eye tracking during locomotion on the HoloLens-2, and marginally, but not significantly, higher eye tracking accuracy on the MagicLeap device.

To ensure the reproducibility of our analysis, we will make the datasets from both studies as well as the source code of EyeTTS publicly accessible to the research community.

2 BACKGROUND AND RELATED WORK

Here we present prior work on stationary eye tracking evaluation, existing calibration methods, and eye tracking during locomotion.

2.1 Eye Tracking Evaluation

Existing work has evaluated the accuracy of eye trackers, but only in a stationary lab setting. Weigle et al. present a protocol to determine the accuracy of an eye tracking device, which consists of users watching the top and bottom edges of an animated bouncing stimulus on a screen [27]. Ehinger et al. present a more rigorous protocol for determining eye tracking accuracy, which consists of ten tasks that present stimuli on a screen and measure accuracy during smooth pursuit, microsaccades, head movements, and other conditions [13]. Aziz and Komogortsev evaluate eye tracking on a HoloLens 2 device using a random saccades task, in which participants fixate on a stimulus presented at random points in the visual field in front of

them [7]. All of these protocols present stimuli on a 2D screen or field of view (FOV), rather than a 3D space, and none of them involve locomotion.

There has also been work into creating generalized packages to evaluate eye trackers. Adhanom et al. present GazeMetrics, a Unity package that includes a calibration procedure to evaluate the accuracy of several supported eye trackers, including the HTC VIVE eye tracker and Pupil Labs add-on eye tracker [6]s. Kapp et al. present the Augmented Reality Eye Tracking Toolkit (ARETT), an eye tracking Unity package for the HoloLens 2 that records eye tracking data and includes an R package for data analysis [17].

In spite of the growing prevalence of head-mounted eye trackers, there is limited work on evaluating eye tracking while moving. The researchers behind ARETT showed that eye tracking accuracy for the HoloLens 2 is lower when a user is walking than when they are standing still [17], but the conditions with which they tested eye tracking while walking were limited to a screen-stabilized saccades procedure. Hausmann et al. examine simultaneous head and eye tracking in a moving setting, but focus on the accuracy of the head and eye tracking data relative to each other, rather than on eye tracking specifically [15].

2.2 Eye Tracking Calibration

Eye tracking calibration involves finding the mapping between the output of an eye tracker and a user's gaze point. There is existing work on finding good calibration methods and functions. Harezlak et al. describe a standard calibration procedure in which users look at defined points on a screen, and a polynomial function is used to map their eye tracking output to those points [14]. Kasprowski et al. find that linear and quadratic functions generally perform better than more complex functions when calibrating eye trackers [18]. Mardanbegi et al. suggest that errors may arise in accuracy when viewed content is at a different depth than the calibration plane [21]. Although existing work thoroughly examines different aspects of the ideal calibration procedure, there has not been any work in finding or generalizing a calibration procedure for eye tracking during locomotion in a 3D environment.

2.3 Eye Tracking during Locomotion

Prior work exists that utilizes eye tracking during locomotion in order to investigate scientific phenomena. Kiefer et al. examine the applications of eye tracking in spatial research and note that eye tracking may provide a valuable tool for researchers looking to investigate how users navigate during locomotion [19]. De Winter et al. demonstrate this by using eye tracking during walking to examine where people focus their attention when walking through a parking garage [10].

Ophthalmologists routinely make use of eye tracking during locomotion to evaluate patients' visual health. Vargas-Martin et al. use mobile eye tracking on patients with severe peripheral visual field construction in order to make recommendations for head-mounted mobility aids [26]. Stuart et al. present an algorithm for detecting saccades during locomotion on the Dikablis eye tracker and test this algorithm on healthy users and users with Parkinson's disease [24]. Vanegas-Arroyave et al. use eye tracking during locomotion to investigate where patients with Parkinson's focus their attention during walking [25].

While eye tracking is increasingly applied in mobile settings, thus coupling eye tracking with ambulatory locomotion, there is still a paucity of studies on how currently available eye trackers are affected by locomotion, especially in the Mixed Reality domain. This work attempts to start closing this knowledge gap and provide evaluation and calibration tools for AR/VR researchers interested in eye tracking while walking.

3 SYSTEM AND IMPLEMENTATION

In this section, we introduce the eye tracking hardware capabilities of the Magic Leap and the HoloLens 2 headsets and discuss the development of the EyeTTS test suite.

3.1 Eye Tracking Hardware

The Magic Leap utilizes one camera per eye for eye tracking [4]. According to Caruso et al, eye gaze information is provided at 60 frames per second using the pupil center and corneal reflections (PCCR) [9, 28]. Based on this information, we anticipate a precision of up to 1° [16]. Although there is limited research on evaluating the accuracy of the Magic Leap device, we can refer to the spatial accuracy of 1.5° for HoloLens 2 [3], a comparable headset that also utilizes PCCR, as a reference point. The MLEyes API offers the gaze position as a 3D point in space, gaze confidence, and a single-direction gaze vector. Additionally, it provides individual gaze confidence, blinking state, eye position, gaze position, and pupil size for both eyes.

The HoloLens 2 integrated eye tracker consists of two infrared cameras, one for each eye, [5] that use corneal reflections to determine a gaze direction. The eye tracking accuracy is reported to be within 1.5 degrees with a sampling rate of 30 Hz [3]. The HoloLens 2 eye tracking API reports gaze as a single direction vector and a gaze origin, without reporting individual eye gaze vectors, a vergence point, blinks, or pupil location.

3.2 Development of the EyeTTS Test Suite

The EyeTTS test suite was initially developed for the Magic Leap headset and later expanded to include support for the HoloLens 2 headset. For the Magic Leap, we utilized the Magic Leap Lumin SDK 0.26, which officially supports Unity editor version 2020.3. We utilized Unity 2020 LTS along with Magic Leap XR Plugin version 6.2.2. For the HoloLens 2, EyeTTS was built using Unity 2019.4.30f1 and the Mixed Reality Toolkit version 2.7.0 with the Windows XR plugin. The package consists of a single Unity scene implemented with Unity World Locking Tools enabled.

Our test suite included tasks that required participants to maintain their gaze on stimuli, which is common in eye gaze studies [11, 22]. The stimuli could be stationary or moving relative to the user and could be presented on a world-stabilized, head-stabilized, or body-stabilized reference frame. This allowed for data collection from conditions representative of various situations in AR.

Each of the tasks involved a hold-and-go movement pattern, where the stimuli followed a seemingly random but predetermined path. These alternating phases were labeled appropriately during data collection to segregate static and dynamic stimulus data points. During the hold phase, the gaze data captured corresponded to fixations, representing static elements within the augmented environment. Conversely, in the go phase, the stimulus observed represents animated and moving elements. The gaze data in the go phase could encompass fixations, smooth pursuit, or saccades, depending on the specific conditions of the task.

We designed the stimuli movement patterns to cover the entire FOV of the user. A frame was placed around the area where the stimulus moved, which was ensured to be within the FOV by making positional and orientational adjustments before starting the task. This frame was not visible during the task.

4 USER STUDY DESIGN

Two separate user studies were carried out to assess the EyeTTS test suite and evaluate the performance of eye tracking on different devices. The initial user study (n=36) was conducted utilizing the Magic Leap headset. The subsequent user study (n=54) was performed using the HoloLens 2 headset, with slight modifications based on feedback and results from the first user study. While in-depth comparisons between the two headsets are limited due to

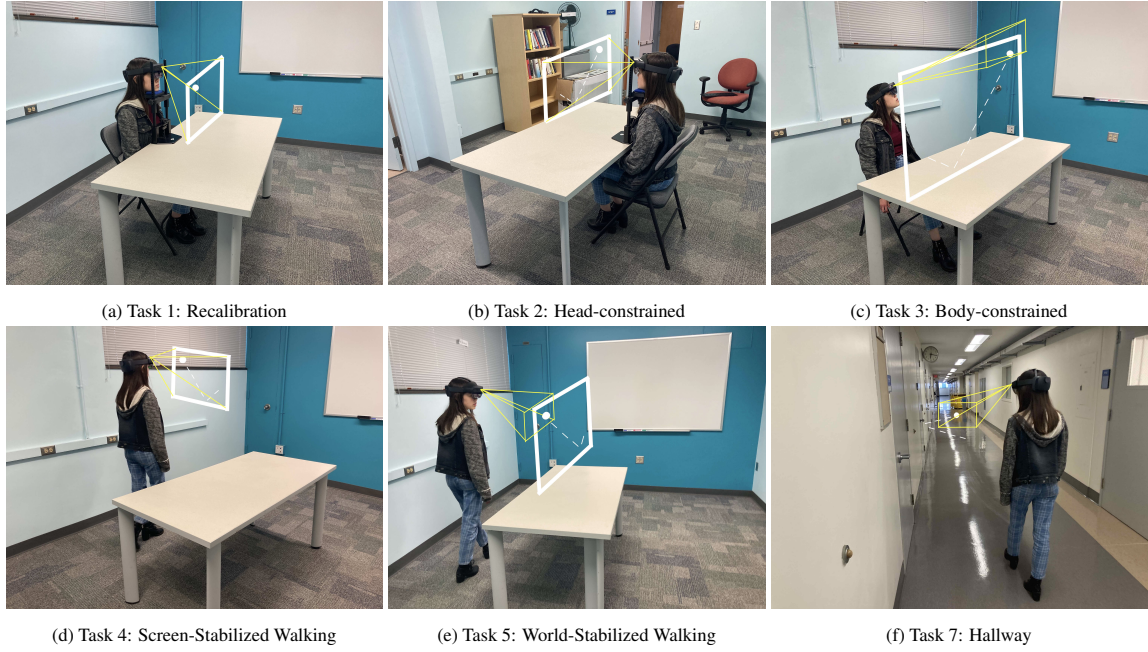


Figure 1: The 6 tasks performed in our user study using EyeTTS on HoloLens 2. AR content is colored in white, while yellow lines approximate the FOV of the headset. The users did not see the frame or path (dashed line) of the target once the experiment began.

the developments made to EyeTTS between the two studies, some general comparisons are made.

4.1 Participants

The first user study on the Magic Leap included a total of 36 university students, with an average age of 22.75 and an equal number of participants identifying as male and female. Out of the total participants, 26 had prior experience with virtual reality (VR) or AR headsets and had previously worn a headset. The experiment consisted of six tasks, and each participant performed each task once.

The second user study on the HoloLens 2 included a total of 54 participants, of which 29 identified as female and 25 identified as male. The participants were recruited from a university campus and had an average age of 22.4 years. Of the participants, 34 had previous experience with AR.

The second experiment was conducted on two HoloLens 2 devices. In order to test for consistency between devices of the same type, half of the participants completed the experiment on one HoloLens (HoloLens A), and the other half completed the experiment on another HoloLens (HoloLens B). The experiment consisted of six tasks, and each participant performed each task twice, in two separate trials, in order to facilitate additional data collection.

4.2 Experiment Procedure

To begin, each participant was asked to read and sign a consent form prior to beginning the experiment. Upon signing the consent form, each participant then underwent the in-built eye calibration procedure built onto the device while keeping their head stationary by resting it on a Tobii Chin Rest. Following the calibration, each participant completed a pre-study survey/questionnaire, providing demographic information and assessing their familiarity with AR. Subsequently, participants engaged in the tasks specific to their study.

4.3 Tasks

Each user study consisted of six tasks from the test suite, with five tasks being identical across platforms to ensure comparability. Each

study included a unique sixth task specifically designed to align with the research objectives and characteristics of the headset platform. To reduce order effects, predefined task orders were generated using the Latin Square method, which is a balanced and counterbalanced design technique. Furthermore, most tasks adhered to the hold-and-go movement pattern for the target, as described in Sect. 3.2.

All tasks featured screen-stabilized, body-stabilized, or world-stabilized content. In user study 1 on the Magic Leap, world-stabilized content was anchored to the real world with images detected by the Magic Leap XR Plugin. For user study 2 on the HoloLens 2, the world-stabilized content was manually anchored to the real world by the researcher prior to the experiment. User study 1 incorporated body-stabilized content with the use of the Magic Leap controller, which had virtual content anchored to it and was strapped to a participant's chest. Due to the lack of an in-built HoloLens 2 controller, user study 2 did not make use of any body-stabilized content.

Several tasks involved participants walking in circles around a table. In order to minimize directional effects in these walking tasks, half of the participants were asked to walk clockwise around the table, while the other half were asked to walk counter-clockwise. Furthermore, in user study 1, participants were not given instructions regarding walking speed, which resulted in different participants walking at different speeds. In order to control walking speed, user study 2 introduced a metronome at 80 bpm for tasks involving walking around a table, which participants were asked to match their footsteps to.

In both user studies, we received participant feedback that the tasks involving locomotion around a table felt unnatural. Upon receiving this feedback and reflecting on these tasks, we realized that these tasks, although involving locomotion, did not reflect common use cases of AR with locomotion. As a result, we added the hallway task in the second study to fill this gap and attempt to create a more natural walking task. We considered the hallway task to be the most representative of various AR scenarios due to its simplicity and resemblance to typical AR interactions.

4.3.1 Task 1: Recalibration

The recalibration task aimed to simulate the eye calibration procedure found in the headsets' built-in systems. Participants were asked to rest their heads on a Tobii Chin Rest in order to keep their heads from moving during the task. During the task, a tracking stimulus appeared at random points in front of the participant, within their entire FOV. The stimulus did not move in between points, and it remained at each point for 2.5 seconds. The AR content was screen-stabilized and programmed to appear 1.25m-1.5m away from the participant's eyes, as recommended by [1]. Participants were asked to look at the stimulus as soon as it appeared without moving their head.

4.3.2 Task 2: Head-Constrained

Task 2 was a head-constrained task with a similar setup to the recalibration task. Participants were asked to rest their heads on a Tobii Chin Rest to minimize head motion. The tracking stimulus moved on preprogrammed paths within a screen-stabilized plane located 1.25m in front of the user. The stimulus took 5 seconds to complete a path and stayed in place for 3 seconds between movements for user study 1 on the Magic Leap. For the second user study on the HoloLens 2, the stimulus only stayed in place for 1.5 seconds between moving paths. The speed of the stimulus during the moving phase consistently measured below $40^\circ/\text{sec}$, remaining within the smooth pursuit limit of $60^\circ/\text{sec}$ [23]. Therefore, we classify this screen-stabilized task as a smooth pursuit task, which is relevant to our precision calculations in Sect. 5.1.

4.3.3 Task 3: Body-Constrained

Task 3 was a body-constrained fixation task, meant to emulate a situation in which a user is stationary in augmented space while observing content that is world-stabilized. Participants could move their heads freely, but were asked to stay seated throughout the task. Furthermore, we requested the participant to follow the target closely and use head movements to always maintain it in the center of their FOV. In this task, the tracking stimulus moved on preprogrammed paths around a world-stabilized plane located in front of the participant. The paths were designed to be sufficiently long enough to require participants to turn their heads in order to observe the stimulus.

4.3.4 Task 4: Screen-Stabilized Walking

Task 4 was an unconstrained task in which participants were asked to walk in circles around a table. The tracking stimulus moved in preprogrammed paths around a screen-stabilized space located 1.25m in front of the participant. Similar to task 2, the head-constrained task, this task was categorized as a smooth pursuit task. However, unlike task 2, it presented the additional challenge of incorporating locomotion while performing eye tracking exercises. In user study 1 on the Magic Leap, participants were free to walk at any speed during this task, but in user study 2 on the HoloLens 2, participants were asked to match their steps to a metronome at 80 bpm.

4.3.5 Task 5: World-Stabilized Walking

Task 5 was an unconstrained task in which participants walked in circles around a table, as shown in Fig. 1e. Unlike in task 4, the screen-stabilized walking task, the tracking stimulus moved in preprogrammed paths around a world-stabilized space located over the table. This forced the participant to turn their head to look at the stimulus while walking. In user study 1 on the Magic Leap, participants were free to walk at any speed during this task, but in user study 2 on the HoloLens 2, participants were asked to match their steps to a metronome at 80 bpm.

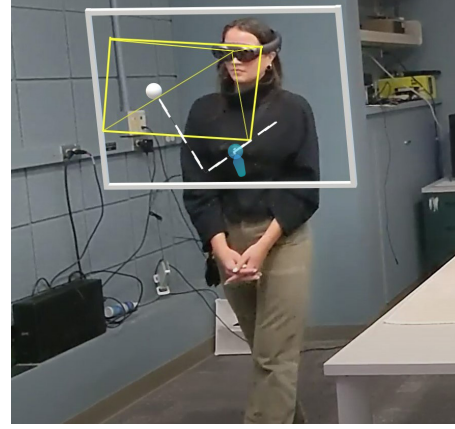


Figure 2: Task 6: Body-Stabilized Walking with the controller highlighted in blue.

4.3.6 Task 6: Body-Stabilized Walking

Fig. 2 shows the body-stabilized walking task, which only occurred in the user study 1 on the Magic Leap. In this task, the target frame was designed to follow and change direction with the user's body. Consequently, this dynamic reference frame led to a greater degree of relative motion between the user and the stimulus.

This task necessitates the use of an accessory with a position and orientation sensor that transmits this information to the headset. For the Magic Leap, the controller was used to constrain the tracking stimulus's movement around it and was affixed to the participant's body using a strap, enabling the creation of a body-stabilized reference. This task can yield valuable insights into eye tracking with respect to a body-stabilized reference frame as utilized by some AR interfaces (e.g. virtual tool belts or personal information landscapes [20]) and can be supported by VR and AR HMDs that offer similar controllers, such as the HTC Vive, Meta Quest Pro, and Samsung HMD Odyssey. However, the task was not executed in our user study involving the HoloLens 2 headset as it lacks any such accessory.

4.3.7 Task 7: Hallway

In task 7 (cf. Fig. 1f), participants were asked to walk in a manner that was meant to be more natural than in tasks 4 and 5, the tasks involving locomotion around a table. Participants were asked to walk down a hallway, following the tracking sphere as it moved back and forth across the hallway in front of them. The participants were not provided with specific instructions regarding walking speed or distance from the stimulus, in order to encourage natural walking. This task only occurred in the second user study on the HoloLens 2, replacing the body-stabilized walking task.

4.4 Data Analysis

4.4.1 Data Preprocessing

When visualizing the data recorded by the HoloLens 2, a latency between the reported head position and the reported eye position was observed. We minimized the latency between the head position and the eye position by shifting the eye position and corresponding eye gaze data backward until the mean euclidean distance between head position and eye position was minimized. This shift was limited to 1.2 seconds. This latency was not observed for data collected on the Magic Leap during user study 1, so this correction was not performed for user study 1.

For correctly analyzing data from task 1, the head-constrained static task, and the static hold phase of all other tasks, we needed to identify periods of stable fixation and eliminate errors due to

saccadic movements. The initial period in the fixations consists of instability due to non-fixational movements [7]. Aziz and Komogortsev [7] determined that by excluding the initial 233ms of fixations, the largest contiguous collection window could be obtained with minimal errors attributed to such instability. We opted to discard the initial 0.5 seconds of data collected each time the tracking sphere changed positions or settled on a new position after movement, while retaining the subsequent 1.5 seconds of data.

4.4.2 Recalibration

Although each participant completed the Magic Leap or HoloLens 2's built-in eye calibration, we wanted to see whether we could improve on these results with further recalibration on the other tasks. To that end, we used linear regression:

$$x' = A_x x + B_x y + C_x, \quad y' = A_y x + B_y y + C_y$$

The coefficients A, B, and C were calculated using the data from whichever task was used for recalibration, such as task 1, and the coefficients were applied to data from other tasks.

4.4.3 Cosine Error

The HoloLens 2 eye tracking API reported eye gaze direction as a 3-dimensional vector. For the Magic Leap, which reported a gaze point rather than a direction, the direction vector was calculated using that gaze point and reported head position. The gaze vector v and the target vector u were used to calculate cosine error using the following formula:

$$\theta = \arccos \frac{\vec{u} \cdot \vec{v}}{||u|| ||v||}$$

Cosine errors for each sample were averaged by participant, and each participant's average cosine error was averaged again to calculate the total average cosine error.

4.4.4 Euclidean Error

In order to convert the 3D gaze direction and target vectors ($v = (v_x, v_y, v_z)$) into 2D screen coordinates, we used the following formulas:

$$x = \arctan\left(\frac{v_x}{v_z}\right), \quad y = \arctan\left(\frac{v_y}{v_z}\right)$$

We then used the 2D screen coordinates to calculate Euclidean error:

$$\theta = \sqrt{(x_{target} - x_{gaze})^2 + (y_{target} - y_{gaze})^2}$$

Euclidean errors for each sample were averaged by participant, and each participant's average error was averaged again to calculate the total average Euclidean error.

4.4.5 Spatial Precision

Spatial precision for transition periods, fixations, and smooth pursuit was calculated using the data from tasks 1 and 2. The following formula was used:

$$\theta_{RMS} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\theta_i)^2}$$

where θ_i is the Euclidean distance between consecutive samples. n represents the number of samples in one fixation or period of movement. Precision for the transition period between fixations was calculated using the first 0.5 seconds after each target movement in task 1. Precision for fixation was calculated using the remainder of the time the target was stationary for each position in task 1. Finally, precision for smooth pursuit was calculated using the periods when

Device	Overall	Transition	Fixation	Smooth Pursuit
Magic Leap	1.5038°	3.6378°	0.4193°	0.4543°
HoloLens 2	0.665°	2.2699°	0.1028°	0.1428°

Table 1: Precision Values for the Magic Leap and HoloLens 2

the target was moving in task 2. Each period of target movement or fixation was considered to be n samples, and the precision value was averaged over one participant. Average precision values for each participant were then averaged to find an overall precision value.

5 RESULTS

Results for both the Magic Leap and the HoloLens 2 study are presented here.

5.1 Spatial Precision

Table 1 shows precision values for the Magic Leap and HoloLens 2 in various conditions taken from both head-constrained tasks. As expected, both headsets exhibit a greater precision during fixation and smooth pursuit than during the transition period between stimulus locations.

5.2 Cosine Error

Fig. 3 shows the graphs of cosine error split by task. Magic Leap data is shown in the top, while HoloLens 2 data is shown on the bottom graph.

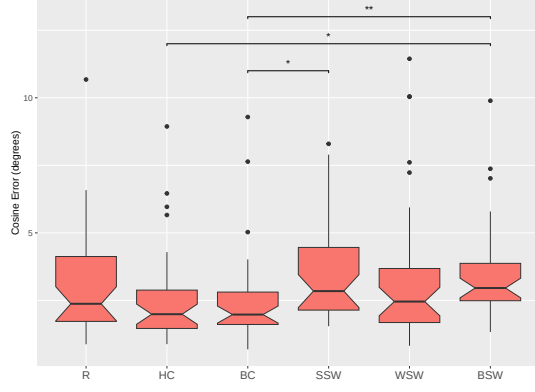
For the Magic Leap, an ART ANOVA with task as an independent variable reveals a significant difference in cosine error between tasks ($F = 11.04, p = 2.95 \times 10^{-9}$). A Bonferroni-corrected Mann-Whitney U test reveals a difference in cosine error between the head-constrained and body-stabilized walking tasks ($p = 0.019$), the body-constrained and the screen-stabilized walking tasks ($p = 0.031$), and the body-constrained and the body-stabilized walking tasks ($p = 0.00012$).

For the HoloLens 2, an ART ANOVA revealed no significant difference in cosine error between HoloLens devices, and a difference in cosine error between trials for the recalibration task only. As a result, for all subsequent analysis, data from both devices and trials was combined into a single dataset, as graphed in Fig. 3. An ART ANOVA with task as an independent variable reveals a significant difference in cosine error between tasks ($F = 1877, p < 2.22 \times 10^{-16}$), and a Bonferroni-corrected Mann-Whitney U test reveals a difference in cosine error between every pairwise combination of tasks.

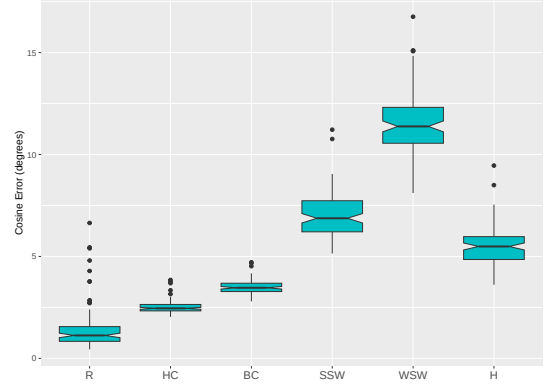
5.3 Magic Leap and HoloLens Comparison

Although both user studies were conducted with the intention of comparing performance on the Magic Leap and HoloLens 2, much more data was collected for the HoloLens 2, with that user study containing more participants and more data gathered per participant. Therefore, it is difficult to determine whether differences in results are entirely due to differences in headset performance, or whether they can be explained by differences in user studies. In order to minimize this difference, 36 participants were randomly selected from the 54 HoloLens 2 participants, and only the data from their first trial was used to generate Fig. 4. In addition, the sixth task of each study (the hallway task and the body-stabilized walking tasks) were eliminated from this comparison, since they were not identical between studies. In this way, we hoped to minimize differences in results due to different amounts of data and more accurately compare Magic Leap and HoloLens 2 performance.

A two-way ART ANOVA with independent variables task and device reveals a significant effect of device on cosine error ($F = 69.997, p = 3.86 \times 10^{-12}$), as well as an interaction between device and task ($F = 239, p < 2.22 \times 10^{-16}$). A Bonferroni-corrected Mann-Whitney U test post-hoc reveals a difference in cosine error



(a) Cosine error per task for the Magic Leap. Notches show 95% confidence interval, * denotes $p < 0.05$, and ** denote $p < 0.01$.



(b) Cosine error per task for the HoloLens 2. Notches show 95% confidence interval. Significance bars omitted as all pairs differ significantly

Figure 3: Cosine error graphs for both the Magic Leap and the HoloLens 2

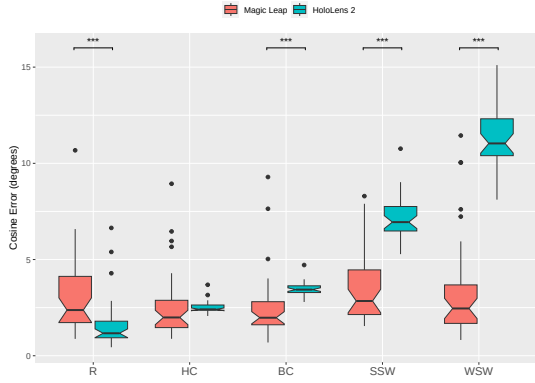


Figure 4: Cosine error for Magic Leap and a subset of HoloLens 2 data plotted side-by-side. Notches denote 95% confidence interval.

between devices for the recalibration task ($p = 5.61 \times 10^{-6}$), the body-constrained task ($p = 1.49 \times 10^{-7}$), the screen-stabilized walking task ($p = 1.23 \times 10^{-9}$), and the world-stabilized walking task ($p = 4.87 \times 10^{-12}$).

5.4 Stimulus Movement

In order to evaluate accuracy for different target movements, we separated the data according to when the tracking stimulus was moving between points and when the tracking stimulus was static. Task 1, the recalibration task, only contained static data and was excluded from this analysis.

Fig. 5 shows graphs of cosine error for the static and moving stimulus for both user studies. For the Magic Leap, an ART ANOVA with independent variables task and movement revealed a significant effect of movement on cosine error ($F = 10.14, p = 0.0016$), as well as an interaction between task and movement ($F = 5.89, p = 1.38 \times 10^{-4}$). A Bonferroni-corrected Mann-Whitney U test revealed a significant effect of stimulus movement for task 4 ($p = 0.0118$). Fig. 5 suggests that cosine error is higher when the target is static for task 4.

For the HoloLens 2, an ART ANOVA with independent variables task and movement reveals a significant effect of target movement on cosine error ($F = 771, p < 2 \times 10^{-16}$), and an interaction between task and movement ($F = 124, p < 2 \times 10^{-16}$). A Bonferroni-corrected Mann-Whitney U test revealed a significant difference in

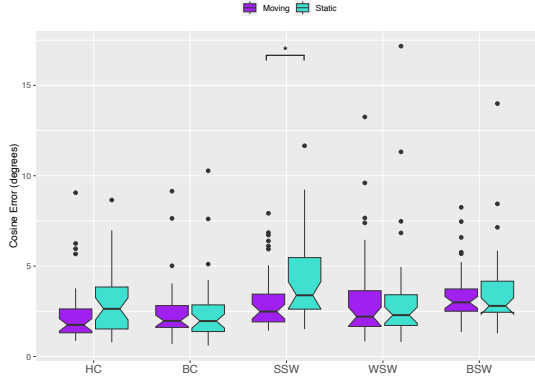
error when the target is moving or static for the head-constrained task ($p = 9.94 \times 10^{-35}$), the body-constrained task ($p = 6.83 \times 10^{-37}$), and the hallway task ($p = 6.46 \times 10^{-23}$). Fig. 5 suggests that cosine error is higher when the target is moving than when it is static for all affected tasks.

5.5 Recalibration

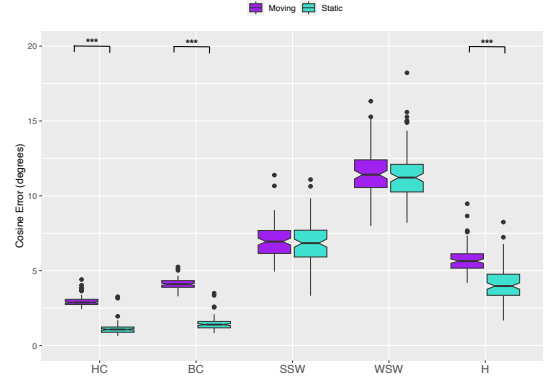
Recalibration was performed on every task in order to find a potential recalibration method suited for eye tracking during locomotion. In general, data from the Magic Leap showed fewer statistically significant differences as a result of recalibration than did data from the HoloLens 2. For the Magic Leap, only recalibration on the head-constrained and world-stabilized walking tasks resulted in a significant difference in a task other than recalibration itself. However, in the case of recalibration on the world-stabilized walking task, only that task showed improvement as a result of recalibration, making every other affected task less accurate.

In order for a task to be a suitable candidate for recalibration, it must both result in improvements and be easy to perform in any situation prior to AR use. Specifically, we are interested in improvements to eye tracking during locomotion, so we decided to prioritize recalibrations that resulted in improvements to the walking tasks over recalibrations that resulted in improvements to the head-constrained or body-constrained tasks. Furthermore, even if recalibrating on a walking task results in improvements to other walking tasks, a walking recalibration task would require a specialized environment for every user and would be impractical for recalibration. This leaves the recalibration, head-constrained, and body-constrained tasks as suitable tasks for recalibration. Looking at the results of those recalibrations, we choose to focus on the body-constrained task for the bulk of our analysis, as it brought about the most consistent improvement in eye tracking performance while walking.

Fig. 6 shows graphs of Euclidean error with device calibration and recalibration using the body-constrained task for both user studies. For the Magic Leap, an ART ANOVA revealed an interaction between recalibration and Euclidean error ($F = 5.02, p = 1.81 \times 10^{-4}$), but no significant effect of recalibration on Euclidean error. A Bonferroni-corrected Mann-Whitney U test revealed a significant difference between device-calibrated and recalibrated error for the body-constrained task ($p = 1.09 \times 10^{-5}$), which makes sense since this was the task that was used for recalibration. Although the other tasks do not show a statistically significant difference, Fig. 6 suggests that the mean error for the recalibrated data is numerically higher for all screen-stabilized tasks, but numerically lower for the other tasks, including those involving walking.



(a) Cosine error for the Magic Leap, separated by stimulus movement. Notches show 95% confidence interval, * denote $p < 0.05$.



(b) Cosine error for the HoloLens 2, separated by stimulus movement. Notches show 95% confidence interval, *** denote $p < 0.001$.

Figure 5: Cosine error graphs for stimulus movement for the Magic Leap and HoloLens 2.

For the HoloLens 2, an ART ANOVA with independent variables task and calibration reveals a significant effect of recalibration on Euclidean error ($F = 234, p < 2.22 \times 10^{-16}$), and an interaction between task and recalibration ($F = 327, p < 2.22 \times 10^{-16}$). A Bonferroni-corrected Mann-Whitney U test revealed a significant difference between device-calibrated and recalibrated error for all tasks ($p_{Task1} = 8.46 \times 10^{-35}, p_{Task2} = 2.99 \times 10^{-34}, p_{Task3} = 1 \times 10^{-29}, p_{Task4} = 5.56 \times 10^{-3}, p_{Task5} = 4.56 \times 10^{-9}, p_{Task6} = 2.51 \times 10^{-11}$). Figure 6 suggests that recalibration may have resulted in a higher Euclidean error for the head-constrained static, head-constrained moving, and screen-stabilized walking tasks, but a lower Euclidean error for all other tasks.

6 DISCUSSION

While we introduced a series of changes to our EyeTTS regime between running our studies on the Magic Leap and HoloLens-2 (increasing the number of participants, introducing repeat trials, enforcing uniform walking speed for circling around the table, new hallway task), we feel that both user studies produced interesting results that point to general insights that could improve eye tracking performance while walking.

For the HoloLens 2, the error between tasks follows a very clear trend in which head-constrained tasks have the lowest error, followed by the body constrained task, then walking tasks. The latency between the head position and the eye position in all walking tasks may have resulted in the lack of accuracy for the walking tasks. Although we attempted to correct this latency by shifting all walking gaze data up to 1.2 seconds post-hoc, it is possible that this correction was not entirely effective in reducing resulting errors. This latency may have been caused by the HoloLens 2 eye tracking sampling rate of 30 Hz, which is lower than that of other headsets (such as the Magic Leap, which has an eye tracking sampling rate of 60 Hz).

Among the walking tasks, the hallway task had the lowest error, followed by the screen-stabilized walking task and the world-stabilized walking task. Several participants reported some dizziness and nausea during the round-table screen-stabilized walking task, and some reported finding the world-stabilized walking task extremely difficult due to the awkwardness of turning their head in a different direction than their walking direction. No participants made any negative comments about the hallway task. Therefore, it is possible that the higher errors in the table walking tasks are due to the difficulty of those tasks and lack of user adherence to the task. It is also possible that the numerous changes in head orientation in those tasks caused the eye tracker to be less accurate.

In contrast, the Magic Leap data displayed a generally higher

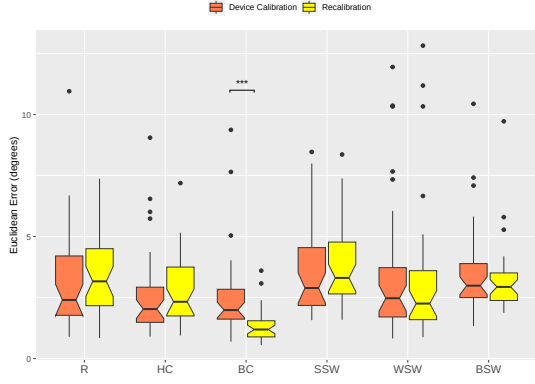
accuracy than the data collected from the HoloLens 2, with a limited loss of accuracy due to locomotion or head motion. The only pairwise combination of tasks that showed any statistically significant differences were the head-constrained task and the body-stabilized walking task, the body-constrained task and the screen-stabilized walking task, and the body-constrained task and the body-stabilized walking task. This suggests that the Magic Leap may have a slight trend towards a loss of accuracy with the addition of ambulatory locomotion, but not with the addition of head motion in the body-constrained case.

The differences between the accuracy of the Magic Leap and the HoloLens 2 may be explained by the fit of the headsets. The Magic Leap rests on the nose, forming a secure fit around a user's head. In contrast, the HoloLens 2 has a visor that can be flipped up, which results in a lack of contact between the display and the face. This means that the visor is held entirely in place by the strap around the forehead, potentially resulting in more headset movement and lower accuracy for tasks with more head and body motion. The lack of stability of the HoloLens 2 on a user's head could explain why the eye tracker is significantly more accurate when the head is kept still, whereas the Magic Leap headset shows more consistent eye tracking accuracy throughout.

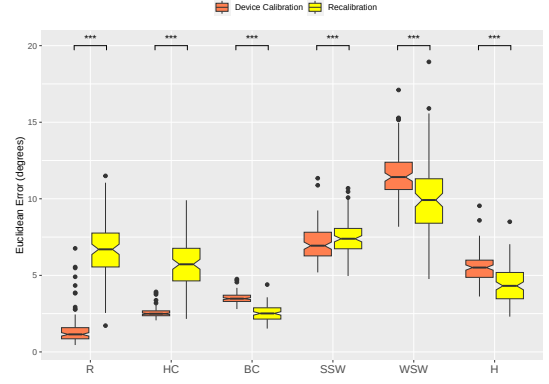
The only task for which it seemed the HoloLens 2 exhibited better eye tracking performance than the Magic Leap was the recalibration task. This can potentially be explained by the larger distance between the display and the user's eyes on the HoloLens 2, which may have allowed the eye tracking cameras to better capture eye gazes at the periphery of the display. Differences in the built-in calibration procedure may also have an effect here.

When it comes to recalibration, the Magic Leap was generally less affected than the HoloLens 2. This may be due to the higher accuracy prior to recalibration, which may suggest that the Magic Leap does not need to be recalibrated as much as the HoloLens 2. For the HoloLens 2, no recalibration task improved performance on every single task. Instead, it was often the case that recalibration on a task that improved conditions during locomotion, such as the body-constrained task, resulted in worse performance during the head-constrained tasks. This may suggest that it is important to consider the end use case of a headset when choosing a recalibration task, and a task with more head motion, such as the body-constrained task, may be an appropriate choice for recalibration for locomotion.

Regarding moving stimuli, the HoloLens 2 data exhibited a clear trend of stimulus movement resulting in lower accuracy compared to the static case for most tasks. Interestingly, this trend is not observed in the Magic Leap data, in which the only task with a statistically



(a) Euclidean error for the Magic Leap. Notches show 95% confidence interval, *** denote $p < 0.001$.



(b) Euclidean error for the HoloLens 2. Notches show 95% confidence interval, *** denote $p < 0.001$.

Figure 6: Euclidean error graphs for Magic Leap and HoloLens 2, recalibrated on the body-constrained task

significant difference between the static and moving target appears to have a higher accuracy when the target is moving. This correlates with the general higher accuracy of the Magic Leap. It is possible that the higher sampling rate of the Magic Leap allows it to handle stimulus and gaze movement more effectively than the HoloLens 2.

We decided to only use the recalibration and head-constrained tasks for precision calculations because they match the existing literature most closely when it comes to precision calculations. Unsurprisingly, transition periods, which are affected more by saccades, produce less precise data than smooth pursuit or fixation periods for both headsets. However, it is worth noting that the HoloLens 2 eye tracking API does not provide a gaze confidence value or any other information that could be used to help determine whether fixation or saccades were occurring. As a result, we approximated the occurrence of saccades as happening 0.5 seconds after the target shifted in task 1, and this may not be an accurate time period. In contrast, the Magic Leap eye tracking API does provide a gaze confidence value, which was used to more accurately determine whether a given sample corresponded to a fixation period.

7 LIMITATIONS AND FUTURE WORK

From the two conducted user studies, one with the Magic Leap headset and the other with the HoloLens 2, we can only compare the two headsets up to a point, as there were variations in study conditions, such as differently controlled factors like walking speed, lighting, and environment. The insights gained from the initial study with Magic Leap helped enhance the robustness of the EyeTTS test suite utilized in the HoloLens 2 study. When aiming to develop a test suite that supports multiple AR platforms universally, we face challenges due to inherent design and feature differences among HMDs. For instance, the Magic Leap had a controller with built-in position and orientation sensors, which facilitated the creation of a body-stabilized reference and gaze confidence and vergence information was not available from the HoloLens-2 API.

In the future, we aim to expand our test suite’s support to include other AR and MR HMDs such as Varjo XR3 and Aero, Meta Quest Pro, Nreal Air, Vuzix Blade, and Apple Vision Pro. Additionally, we plan to test eye tracking on VR devices like HTC Vive Pro 2 and Sony PS VR2, which afford some mobility. Our long-term goal is to develop a universal API that can seamlessly integrate with popular 3D development platforms like Unity, offering a package that can be utilized across multiple platforms. However, this endeavor poses challenges as it naturally veers towards the lowest common denominator of eye tracking capabilities and may require collaboration with manufacturers whose technologies may be proprietary and not open

source, adding complexity to the low-level programming involved.

An additional limitation lies in the tasks themselves. With our existing setup, there was no way to verify that participants completed the tasks correctly. A possible way to rectify this in the future would be to include a behavioral measurement to check a user’s attention. For instance, this could involve periodically briefly changing the color of the tracking sphere and asking the user to click a clicker every time this occurs. Such a behavioral measurement would help verify that a user is paying attention to the tracking sphere and performing the task correctly, and data from users for whom this is not the case could be discarded. The absence of such a behavioral measure necessitates us to make the assumption that the participants maintained their gaze on the target consistently and sincerely throughout the eye tracking task.

8 CONCLUSION

In this work we developed the Eye Tracking Test Suite, EyeTTS, which is designed to test the eye tracking accuracy of an AR device in various user scenarios. The included tests account for when a user is sitting with their head still, sitting with their head freely moving, or walking, as well as whether a user is viewing screen-stabilized, body-stabilized, or world-stabilized content. With this test suite, we conducted user studies on the Magic Leap and HoloLens 2 to analyze their eye tracking performance in different conditions.

Our analysis revealed overall lower eye tracking accuracy while walking compared to stationary poses. Between the two tested headsets, we observed significant differences, especially while walking, which may be explained by different fit and tightness of the headset in front of the eyes. We demonstrated that we can enact improvements to eye tracking performance while walking when a calibration technique involving stipulated head motion is employed.

Our work emphasizes the importance of considering eye tracking accuracy in different scenarios, rather than assuming stationary conditions. Our testing of various recalibration methods fails to reveal a single recalibration method that is suitable for all use cases, which suggests that it is important to consider the end use of a device when choosing a recalibration method.

Overall, our work provides future researchers with a means of investigating the accuracy of the eye tracker of their AR device prior to use. This may allow researchers to gain valuable insights, improving the performance of their eye tracker in mobile mixed reality applications.

REFERENCES

- [1] Microsoft HoloLens Comfort guidelines. <https://learn.microsoft.com/en-us/windows/mixed-reality/design/comfort>, 10 2021.
- [2] Apple Vision Pro, 2023. <https://www.apple.com/apple-vision-pro/>.
- [3] Eye Tracking on HoloLens 2. <https://learn.microsoft.com/en-us/windows/mixed-reality/design/eye-tracking>, 03 2023.
- [4] Magic Leap 1 vs Magic Leap 2: The key differences, 03 2023. <https://vr.x.vr-expert.com/magic-leap-1-vs-magic-leap-2-the-key-differences/>.
- [5] Microsoft hololens - overview, features, and specs. <https://www.microsoft.com/en-us/hololens/hardware>, 2023.
- [6] I. B. Adhanom, S. C. Lee, E. Folmer, and P. MacNeilage. GazeMetrics: An open-source tool for measuring the data quality of HMD-based eye trackers. *Proc. Eye Track. Res. Appl. Symp.*, 2020, June 2020.
- [7] S. Aziz and O. Komogortsev. An assessment of the eye tracking signal quality captured in the HoloLens 2. In *2022 Symposium on Eye Tracking Research and Applications*. ACM, jun 2022. doi: 10.1145/3517031.3529626
- [8] B. T. Carter and S. G. Luke. Best practices in eye tracking research. *International Journal of Psychophysiology*, 155:49–62, 2020. doi: 10.1016/j.ijpsycho.2020.05.010
- [9] T. J. Caruso, O. Hess, K. Roy, E. Wang, S. Rodriguez, C. Palivathukal, and N. Haber. Integrated eye tracking on magic leap one during augmented reality medical simulation: a technical report. *BMJ Simulation Technology Enhanced Learning*, 7(5):431–434, Feb. 2021.
- [10] J. de Winter, P. Bazilinsky, D. Wesdorp, V. de Vlam, B. Hopmans, J. Visscher, and D. Dodou. How do pedestrians distribute their visual attention when walking through a parking garage? an eye-tracking study. *Ergonomics*, 64(6):793–805, June 2021.
- [11] A. T. Duchowski. A breadth-first survey of eye-tracking applications. *Behav. Res. Methods Instrum. Comput.*, 34(4):455–470, Nov. 2002. doi: 10.3758/BF03195475
- [12] A. T. Duchowski. *Eye tracking methodology: Theory and practice*. Springer, 2017.
- [13] B. V. Ehinger, K. Groß, I. Ibs, and P. König. A new comprehensive eye-tracking test battery concurrently evaluating the pupil labs glasses and the EyeLink 1000. *PeerJ*, 7(e7086):e7086, July 2019.
- [14] K. Harezlak, P. Kasprowski, and M. Stasch. Towards accurate eye tracker calibration – methods and procedures. *Procedia Computer Science*, 35:1073–1081, 2014. Knowledge-Based and Intelligent Information Engineering Systems 18th Annual Conference, KES-2014 Gdynia, Poland, September 2014 Proceedings. doi: 10.1016/j.procs.2014.08.194
- [15] P. Hausamann, C. Sinnott, and P. R. MacNeilage. Positional head-eye tracking outside the lab: an open-source solution. In *ACM Symposium on Eye Tracking Research and Applications*. ACM, New York, NY, USA, June 2020.
- [16] C. Jian-nan, Z. Peng-yi, Z. Si-yi, Z. Chuang, and H. Ying. Key techniques of eye gaze tracking based on pupil corneal reflection. In *2009 WRI Global Congress on Intelligent Systems*, vol. 2, pp. 133–138, 2009. doi: 10.1109/GCIS.2009.338
- [17] S. Kapp, M. Barz, S. Mukhametov, D. Sonntag, and J. Kuhn. Arett: Augmented reality eye tracking toolkit for head mounted displays. *Sensors*, 21(6), 2021. doi: 10.3390/s21062234
- [18] P. Kasprowski, K. Harezlak, and M. Stasch. Guidelines for the eye tracker calibration using points of regard. In *Advances in Intelligent Systems and Computing*, Advances in intelligent systems and computing, pp. 225–236. Springer International Publishing, Cham, 2014.
- [19] P. Kiefer, I. Giannopoulos, M. Raubal, and A. Duchowski. Eye tracking for spatial research: Cognition, computation, challenges. *Spat. Cogn. Comput.*, 17(1-2):1–19, Jan. 2017.
- [20] W. S. Lages and D. A. Bowman. Walking with adaptive augmented reality workspaces: design and usage patterns. In *Proceedings of the 24th International Conference on Intelligent User Interfaces*, pp. 356–366, 2019.
- [21] D. Mardanbegi and D. Hansen. Parallax error in the monocular head-mounted eye trackers. pp. 689–694, 09 2012. doi: 10.1145/2370216.2370366
- [22] A. Poole and L. Ball. *Eye tracking in human-computer interaction and usability research: Current status and future prospects*, pp. 211–219. 01 2006.
- [23] L. Schalén. Quantification of tracking eye movements in normal subjects. *Acta Otolaryngol.*, 90(1-6):404–413, Jan. 1980. doi: 10.3109/00016488009131742
- [24] S. Stuart, B. Galna, S. Lord, L. Rochester, and A. Godfrey. Quantifying saccades while walking: Validity of a novel velocity-based algorithm for mobile eye tracking. In *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, pp. 5739–5742, 2014. doi: 10.1109/EMBC.2014.6944931
- [25] N. Vanegas-Arroyave, D. F. Chen, P. M. Lauro, G. Norato, C. Lungu, and M. Hallett. Where do parkinson’s disease patients look while walking? *Mov. Disord.*, 37(4):864–869, Apr. 2022.
- [26] F. Vargas-Martín and E. Peli. Eye movements of patients with tunnel vision while walking. *Invest. Ophthalmol. Vis. Sci.*, 47(12):5295–5302, Dec. 2006.
- [27] C. Weigle and D. Banks. Analysis of eye-tracking experiments performed on a tobi t60. 01 2008. doi: 10.1117/12.768424
- [28] Z. Wu, S. Rajendran, T. As, J. Zimmermann, V. Badrinarayanan, and A. Rabinovich. Magiceyes: A large scale eye gaze estimation dataset for mixed reality, 03 2020.